

# Nikhil Ravi

M.Sc. Autonomous Systems | Software Developer

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## PROFILE SUMMARY

Master's student in Autonomous Systems at Hochschule Bonn-Rhein-Sieg with hands-on experience owning software from design through deployment and operations. Background spans distributed backend systems, cloud infrastructure, and CI/CD-driven development for production robotic platforms, alongside a strong foundation in data structures, algorithms, and object-oriented design. Experienced with GenAI developer tooling and comfortable operating in fast-paced, agile, cross-disciplinary teams.

## TECHNICAL SKILLS

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|-----------------------------------------|-------------------------------------------------------------------------------------|
| <b>Programming Languages:</b>           | Python, C++17, Go, C#, SQL, Bash                                                    |
| <b>Cloud &amp; Distributed Systems:</b> | AWS (S3), Redis, Real-Time Telemetry Pipelines, Microservices                       |
| <b>Software Engineering:</b>            | REST API Development, FastAPI, Flask, Software Architecture, Object-Oriented Design |
| <b>Data &amp; Databases:</b>            | PostgreSQL, SQLite, InfluxDB, Data Modeling, ETL/ELT Pipelines                      |
| <b>DevOps &amp; CI/CD:</b>              | Git, GitHub Actions (CI/CD), Docker, Linux, CMake, Protocol Buffers                 |
| <b>AI/ML &amp; GenAI Tools:</b>         | PyTorch, Transformers, LangChain, LlamaIndex, Scikit-Learn, NLP                     |
| <b>Computer Vision &amp; Robotics:</b>  | ROS2, Gazebo, SLAM, Object Detection, Sensor Fusion, Navigation Stack               |

## PROFESSIONAL EXPERIENCE

**Teaching & Research Assistant** — *Hochschule Bonn-Rhein-Sieg* Bonn, Germany | **May 2026 – Present**

- Own end-to-end design and development of practical coursework for the Autonomous Robotics curriculum, from initial design through delivery and iteration, focused on ROS2 and Gazebo simulation for robotic perception and control.
- Administer the departmental GitHub organization, maintaining repositories and version-control workflows and streamlining assignment distribution and code submission for the student cohort.
- Collaborate cross-functionally with faculty to design lecture materials and deliver structured, code-level feedback on weekly assignments.

**IoT Developer & Software Engineer** — *Flexli Technologies* Bengaluru, India | **Apr 2023 – Feb 2025**

- Designed and operated real-time, distributed telemetry pipelines using MQTT for autonomous robot monitoring, processing streaming sensor data across multiple platforms at scale.
- Set up AWS S3 for durable robot log storage and ran a Redis server as a fast-access log store, building a CRON-based monitoring job to proactively detect and flag errors in production supporting operational correctness for live robotic deployments.
- Developed backend logging servers in Python and Go, improving data storage efficiency by 40% through query optimization and database indexing.
- Built scalable, fault-tolerant data ingestion architectures with structured error handling for production robotic deployments, and implemented GitHub Actions CI/CD pipelines to automate testing and deployment for embedded and backend components.
- Programmed ESP32, STM32, and Raspberry Pi in C++ for sensor integration and autonomous robot behavior, including sensor fusion for state estimation.

**Data Analyst Associate** — *Redwood Algorithms* Bengaluru, India | **Jan 2023 – Mar 2023**

- Automated ETL workflows in Python, reducing manual data processing time and improving pipeline reliability through validation and quality checks.
- Built dashboards and analytical reports from large datasets using PostgreSQL and complex SQL queries for business intelligence use cases.

**Data Engineer Intern** — *Empower COE* Bengaluru, India | **May 2022 – Nov 2022**

- Built REST API data ingestion pipelines and prepared structured datasets for NLP and machine learning model training.
- Benchmarked API performance using Postman and optimized data workflows, achieving measurable reductions in response latency.
- Implemented data validation and cleansing procedures to ensure high-quality training data for downstream ML models.

## KEY PROJECTS

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- Toyota Human Support Robot (HSR) — Vision & Navigation Integration**2025
- Developed ROS2-based perception and navigation modules integrating SLAM, object detection, and manipulation for a distributed robotic system; validated in Gazebo simulation and real-world environments.
- Autonomous Navigation Framework**2025
- Built a ROS2 navigation stack with A\* and Dijkstra path planners, obstacle avoidance, and sensor fusion; quantitatively evaluated performance in Gazebo simulation.
- Full-Stack ERP System**2024
- Designed and built a modular ERP system with a FastAPI backend, normalized PostgreSQL schema, and React frontend, implementing role-based access control, inventory management, and real-time dashboard analytics end-to-end ownership from architecture through deployment.
- Automatic Grading System using RoBERTa**2024
- Fine-tuned a RoBERTa transformer in PyTorch on university exam data to evaluate written answers against grading rubrics; analyzed prediction quality against instructor-assigned scores.
- Data Processing Pipeline**2024
- Built end-to-end Python ETL pipelines with incremental loading, data validation, and optimized PostgreSQL queries, improving throughput for large-scale analytical workloads.

## COMPETITIONS & HACKATHONS

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- Junction 2025, Espoo — 1st Place**2025
- Built an agentic AI system leveraging GenAI tooling for electrical grid management, using digital identities to optimize EV energy consumption and improve grid transparency across urban infrastructure.
- Intrinsic AI Robotics Challenge**2026
- Developed an imitation learning policy using Action Chunking Transformers for robotic manipulation, trained on expert demonstrations and evaluated across multiple task configurations.

## EDUCATION

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- M.Sc. Autonomous Systems — Hochschule Bonn-Rhein-Sieg**Sankt Augustin, Germany | 2025 – Present
- Expected Graduation: Fall 2027
  - Relevant Coursework: Machine Learning, Mobile Robotics, Computer Vision, Data Processing & Algorithms, Software Engineering
- B.E. Mechatronics Engineering — New Horizon College of Engineering**Bengaluru, India | 2017 – 2021
- Relevant Coursework: Software Engineering, Machine Learning, Embedded Systems, IoT, Data Structures, Database Management

## LANGUAGES

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English (C1) | German (A2, Actively improving) | Hindi (Fluent)